Philosopher's Stone: Architecture, Scientific Potential, and Validation Program for the Epistemic Octahedron System

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Abstract

This paper is a companion to the existing graph paper on the Epistemic Octahedron. That companion paper already defines the graph itself: the octahedral surface rule, the axis meanings, the distinction between pre-philosophical nullity and epistemic collapse, and the claim that objective peak philosophical maturity is represented by the upper vertex. The present paper takes those graph semantics as fixed and studies the wider system built around them.

The repository implements a full interpretation pipeline rather than a static visualization. A user enters text. A constrained language-model contract extracts structured evidence. A deterministic profiler converts that evidence into semantic axis values and persistent gate states. The semantic triple is then projected onto the octahedral surface and rendered in an interactive visualizer that preserves geometric validity. This architecture matters because it separates theory from instrumentation and probabilistic extraction from deterministic scoring.

The strongest claim this paper makes is narrow but important. The graph can reasonably be treated as the final theoretical core of the project, while the current contract, signal ontology, weighting scheme, tie-break rules, canon update logic, and gate dynamics should be treated as a provisional measurement specification. That framing protects the theory from being confused with one temporary implementation, while still allowing the implementation to be audited, benchmarked, criticized, and validated as a serious research object.

The paper therefore does three jobs. First, it documents the current architecture and the measurement logic visible in the code. Second, it explains the scientific status of each layer: final theory, provisional operationalization, and interface. Third, it proposes a concrete validation rubric for model selection, inter-rater benchmarking, longitudinal study, and predictive testing. On that basis, the project can be described neither as mere aesthetic metaphor nor as already validated psychometrics, but as a structured candidate instrument with unusually good auditability for this class of worldview modeling.

1 Scope and framing

This manuscript starts from the graph paper, not from scratch. The graph paper already argues for the final semantics of the Epistemic Octahedron: surface positions satisfy $|x| + |y| + |z| = 1$, the origin is a reserved pre-philosophical null state, the lower vertex is epistemic collapse, the upper vertex is objective peak philosophical maturity, and the horizontal axes encode empathy versus practicality and wisdom versus knowledge. Those claims belong to the theory of the graph.

What this manuscript adds is a system paper. It asks what the repository is actually doing, which parts are fixed theory and which parts are still adjustable, what makes the architecture scientifically promising, and what would have to be shown before the profiler could be called a validated instrument rather than a well-specified prototype.

That distinction is the right one for this project. A geometric theory can be coherent before its measurement instrument is mature. In the same way, an implementation can be clever without yet earning scientific trust. The current codebase sits between those two states. It is already specific enough to audit, but it is not yet empirically settled.

2 System overview

At the repository level, Philosopher's Stone is best understood as a layered browser-native pipeline. The main page acts as a workspace for text input, canon editing, packet generation, profile import and export, and display of compiled outputs. The contract layer defines what an LLM may extract. The profiler layer computes the actual semantic values and final point. The visualizer layer renders and edits points on the octahedron while preserving the surface rule. A lightweight avatar layer adds a human-readable region label but does not belong to the scientific core.

In functional terms, the current system has the following dataflow:

1. User text is entered as worldview text, profile text, or a fragment of philosophical language.
2. The workspace builds an LLM packet from the new text, the current canon, the current suggested canon, and profiler memory.
3. The LLM returns structured evidence under a fixed JSON contract.
4. The deterministic profiler normalizes the payload, updates persistent state, aggregates axis evidence, computes semantic values, and projects them to (x, y, z).
5. The visualizer renders the result and can also accept direct manipulation while remaining on the octahedral surface.

This split is the main architectural strength of the project. The language model does not directly decide the final plot. It extracts evidence. The profiler, not the LLM, owns the final semantics and the final projection.

3 Architectural layers

3.1 Workspace and orchestration

The workspace layer, represented by the main interface and its application logic, is not passive glue. It performs several substantive jobs.

It stores profile text, LLM output, canon state, selected avatar, compile history, and math-display state. It builds the outgoing packet that the user can copy into a language model. It keeps separate

lists for current canon and suggested optimization. It can export and import profile state. It can rebuild the profiler state from prior compiled payloads. It can also post finalized profile data into the visualizer.

This matters scientifically because the system is already set up for cumulative profiling rather than only one-shot scoring. The presence of canon memory, gate memory, and compiled history means the repository is implicitly designed for diachronic use. That fits the graph paper's emphasis on movement and development over time.

3.2 Interpreter contract

The contract layer is the epistemic constitution of the system. In the current code, the core contract states that the LLM is an extractor and canon optimizer only, and that it must not compute final scores, maturity percentages, or final coordinates. That one choice does more for auditability than any later formula.

The contract requires the model to return structured JSON containing at least the following categories:

- analysis scope and scope strength,
- statement modes,
- local extraction such as principles, boundaries, values, tradeoffs, and contradictions,
- axis events for the empathy-practicality axis and the wisdom-knowledge axis,
- positive and negative local signals for epistemic stability,
- triggered gate events for six named meta-epistemic gates,
- profile update signals, and
- suggested canon optimization.

The contract also includes two scientifically useful restrictions. First, it says silence is neutral, so the model must not fail a gate merely because a short text did not explicitly show it. Second, it says the model should under-call rather than over-call, which is the right bias in an evidence-extraction setting.

The current contract version is strong enough to count as a typed interface between probabilistic interpretation and deterministic scoring. That is a serious advance over a single opaque prompt that directly outputs coordinates.

3.3 Deterministic profiler

The profiler is the actual measurement engine. It normalizes heterogeneous payloads, merges profile state, updates gate scores, aggregates axis evidence, computes semantic values, produces supporting notes, and projects the final point. In the current code it identifies itself as `epistemic_octahedron_profiler_v6`, which is useful because versioned profiler outputs are easier to audit longitudinally.

The profiler maintains three kinds of durable state:

1. entry history,
2. gate state for the six meta-epistemic gates,
3. profile memory in the form of core principles, core boundaries, meta-epistemic markers, and risk notes.

This is important because the system is not only classifying the latest text fragment. It is building a persistent philosophical profile with memory and revision pressure.

3.4 Visualizer

The visualizer is more disciplined than a typical 3D demonstration. It preserves the surface rule under multiple interaction modes: slider movement, coordinate editing, pointer dragging, view rotation, and external profile updates. It includes explicit projection to the octahedron surface, surface-normal logic for the pointer marker, top-view snapping, coordinate formatting, label placement, and internal self-tests.

That geometric discipline matters. If the graph semantics are meant to be exact, then the user interface should not quietly allow impossible points or inconsistent inverse mappings. The current visualizer shows real care in this area.

3.5 Avatar layer

The avatar layer maps broad spatial regions to named characters such as *steady*, *feral*, *kind*, *gritty*, *sage*, and *geeky*. This layer is useful as interface language, but it should remain explicitly non-scientific. It is a semantic overlay, not evidence.

4 Measurement logic

4.1 Semantics first, coordinates second

The current system does not compute coordinates directly from raw text. It first computes a semantic triple

$$(a, s, b) \in [-1, 1]^3,$$

where a is the empathy-practicality semantic value, s is the epistemic-stability estimate, and b is the wisdom-knowledge semantic value.

Only after those semantic values are computed does the profiler project them onto the graph:

$$(x, y, z) = \frac{(a, s, b)}{|a| + |s| + |b|},$$

when the denominator is nonzero. If the semantic magnitude is zero, the system uses the reserved null state

$$(x, y, z) = (0, 0, 0).$$

This is exactly the right structure for the project. The graph theory remains clean, and the measurement layer remains separable from the rendering layer. It also means the system respects the graph paper's distinction between the origin and active surface points.

4.2 Lateral-axis aggregation

For each lateral axis, the profiler accumulates directional pole evidence, integration events, and compact signals if present. The current code uses a moderation logic in which integration dampens overdrawn polarization without automatically erasing directional lean.

A compact description of the current rule is:

$$r_u = \mathrm{clamp}\left(\frac{(P_u - N_u)(1 - \rho_u \lambda_I)}{\sigma_u}, -1, 1\right),$$

where P_u is positive-pole mass, N_u is negative-pole mass, ρ_u is the integration ratio, λ_I is integration influence, and σ_u is saturation.

The logic is reasonable. Pole evidence says where the text leans. Integration says whether the text manages tension rather than simply loading one side. In other words, integration moderates extremity but does not erase content.

4.3 Epistemic-stability aggregation

The y-axis is richer than the lateral axes because it combines local signals, contradiction pressure, and persistent gate state.

A compact statement of the current logic is:

$$y_{\text{local}} = \mathrm{clamp}\left(\frac{P_Y - N_Y - \alpha C}{\sigma_Y}, -1, 1\right),$$

followed by

$$y_{\text{estimate}} = \mathrm{clamp}\left(y_{\text{local}} + \beta\overline{G^+} - \gamma\overline{G^-}, -1, 1\right).$$

Here P_Y is weighted positive local signal mass, N_Y is weighted negative signal mass, C is contradiction penalty, and $\overline{G^+}, \overline{G^-}$ are weighted positive and negative persistent gate means.

The profiler also computes a distinct coverage quantity,

$$y_{\text{coverage}} = \frac{\sum w_g^{\text{covered}}}{\sum w_g^{\text{all}}},$$

which estimates how much of the gate ontology has actually been touched by evidence.

This split between estimate and coverage is one of the best scientific choices in the repository. A strong score based on thin evidence should not be treated the same way as a strong score supported

by broader gate exposure.

4.4 Gate dynamics

The six current gates are counter-consideration, non-strawman, self-correction, contradiction handling, reality contact, and non-self-sealing. Each gate stores a score, a status, counts of positive and negative events, and the last evidence span.

New gate events update the stored score with a memory-sensitive rule. Reinforcing events dampen as the gate saturates in one direction. Opposing events apply stronger reversal pressure. This is a good design choice because it discourages cheap farming and turns the gates into actual profile memory.

Just as important is the contract rule that silence is neutral. A text fragment should not get punished for failing to display every possible virtue. That rule is essential if the system is meant to process narrow fragments as well as full imports.

4.5 Tie-breaking and lexical leakage

The profiler contains a fallback tie-break for under-specified lateral cases where both poles arrive with equal default weights and no explicit balance signal. In such cases it searches local text for weighted keywords to nudge the tie.

This is useful as a practical safeguard against dead symmetry caused by weak extraction, but it is also a clear scientific risk point. Any lexical tie-break introduces dependence on wording, vocabulary, and training-style artifacts. This feature should therefore be treated as provisional and stress-tested heavily during validation.

5 What is final and what is provisional

The single biggest conceptual cleanup needed for this project is a strict distinction between final theory and provisional operationalization.

5.1 Reasonably final components

The following components can reasonably be treated as final or near-final for the current phase of the project:

- the octahedral surface rule $|x| + |y| + |z| = 1$ for active positions,
- the reserved origin as pre-philosophical null state,
- the upper vertex as objective peak philosophical maturity,
- the lower vertex as epistemic collapse,
- the axis meanings: empathy versus practicality, wisdom versus knowledge, and negative versus positive epistemic stability,

- the general claim that the graph is diachronic and can track movement over time.

These belong to the graph theory itself.

5.2 Provisional components

The following components should be treated as adjustable measurement design rather than settled truth:

- the wording of the contract,
- the exact ontology of local signals,
- the current six-gate set and their relative weights,
- scope weights, strength weights, and saturation constants,
- contradiction penalties and positive versus negative gate influence,
- lexical tie-break rules,
- canon merge rules and memory caps,
- avatar mapping and other UI semantics.

This distinction is not a weakness. It is the correct scientific posture. The graph can stay stable while the instrument around it improves.

6 Scientific potential

The system has real scientific potential for five reasons.

First, it is auditable. The LLM output is structured. The profiler is deterministic. The projection is explicit. The final artifact includes formulas, diagnostics, and source counts.

Second, it is modular. Extraction quality, gate design, contradiction handling, and visualization fidelity can all be tested independently rather than hidden inside one model output.

Third, it is longitudinal by design. The system stores canon and gate state across entries, which makes developmental tracking possible in principle.

Fourth, it distinguishes score from coverage. That is rare and valuable.

Fifth, it preserves theory-implementation separation. The graph semantics do not have to rise or fall with one temporary prompt or weighting scheme.

The potential applications remain broad and align with the graph paper: worldview research, educational development, reflective counseling tools, philosophical anthropology, institutional analysis, and AI systems that need richer human-model structure than preference lists alone.

7 Threats to validity

The project is promising, but several risks are obvious.

7.1 Construct leakage

The current extraction categories, tie-break terms, and signal labels could smuggle in assumptions that are not actually part of the graph theory. If the measurement layer grows too idiosyncratic, the profiler may start tracking the language of the contract more than the intended constructs.

7.2 Prompt and model sensitivity

Two LLMs may produce different extraction payloads from the same text even when given the same contract. That threatens reliability. Model selection therefore cannot be casual.

7.3 Lexical bias

Any keyword-based tie-break system risks over-reading certain words and under-reading semantically equivalent wording. That feature should be treated as a fallback heuristic, not as a truth mechanism.

7.4 Granularity mismatch

A narrow text fragment may not express enough content to justify strong longitudinal updates. The contract partially protects against this by classifying scope and by treating silence as neutral, but the risk remains.

7.5 Normative inflation

Because the theory includes a normative summit, there is a temptation to overclaim what current scores mean. The project should resist that temptation. The theory may define a peak, but the instrument still has to earn trust in how it estimates movement toward or away from that peak.

8 A concrete rubric for extractor-model selection

Because the profiler depends on upstream extraction quality, model selection should be explicit. The right benchmark is not a generic chatbot leaderboard. The right benchmark is a repository-specific evidence-extraction benchmark built around the contract.

8.1 Benchmark unit

The benchmark unit should be a single input text plus a gold or adjudicated structured extraction. The benchmark should include at least three scope classes: thought, stance, and worldview fragment, with full-profile imports added later when enough data exists.

8.2 Scoring dimensions

A practical rubric for extractor selection is shown in Table **??**.

Table 1: Proposed extractor-model benchmark rubric

Dimension	Weight	What counts as good performance
Contract compliance	0.20	Returns valid JSON, respects required fields, does not compute final scores or coordinates, does not hallucinate forbidden outputs.
Scope classification	0.10	Correctly classifies thought, stance, worldview fragment, or full-profile import with reasonable strength labeling.
Principle and boundary extraction	0.15	Extracts portable principles and boundaries without parroting surface wording or inventing extra canon.
Axis-event precision	0.15	Correctly identifies x-axis and z-axis directional evidence and integration events with low over-calling.
Gate-event precision	0.15	Fires the correct gate events only when text supports them and respects the neutrality-of-silence rule.
Contradiction handling	0.10	Detects genuine contradiction or tension without turning normal tradeoffs into contradictions.
Evidence-span quality	0.05	Quotes or points to concise supporting text spans that actually justify the extraction.
Canon-optimization quality	0.05	Suggests compression or sharpening of canon wording without semantic drift.
Cross-run stability	0.05	Produces similar extractions across repeated runs on the same input.
Adversarial robustness	0.05	Resists rhetorical noise, decorative language, and leading wording that should not change the extraction materially.

A conservative acceptance rule would require high contract compliance, high gate-event precision, and low over-calling before any model is allowed into the pipeline. A model that sounds insightful but violates the contract should be rejected.

8.3 Why this benchmark should be single-turn first

Single-turn extraction is the right first benchmark because the current architecture already gives the profiler persistent memory. The extractor does not need long autobiographical context to do its

job. It needs to read one text accurately under one contract. That makes benchmarking cleaner and cheaper.

9 A concrete rubric for scientific validation

Once extractor quality is under control, the next step is instrument validation. Table **??** gives a practical staged rubric.

Table 2: Proposed validation rubric for the full system

Stage	Test	Priority	Success criterion
Content validity	Expert review of graph semantics and extraction ontology	High	Reviewers agree that the categories actually reflect the intended constructs and do not collapse distinct lower-half conditions.
Inter-rater agreement	Humans independently annotate benchmark texts under the same rubric	High	Acceptable agreement on principles, boundaries, gate events, and broad axis direction.
Model-to-human agreement	Compare LLM extraction with adjudicated human extractions	High	Strong agreement on gate precision and useful agreement on axis direction.
Test-retest reliability	Repeat the same input across runs, model versions, and prompt copies	High	Low variance in extracted structure and small coordinate drift.
Sensitivity analysis	Perturb weights, tie-break rules, and wording slightly	Medium	Coordinates remain directionally stable unless the input genuinely changes.
Longitudinal coherence	Feed sequential texts from the same person or corpus	High	Movement over time is interpretable rather than noisy or arbitrary.
Known-groups validity	Compare corpora designed to differ in stability, contradiction handling, or reality contact	Medium	Groups separate in the expected direction without crude stereotype collapse.
Predictive validity	Test whether plots forecast later coherence, revision quality, or contradiction resolution	High	Position or movement predicts downstream epistemic behavior better than chance and better than naive baselines.

Stage	Test	Priority	Success criterion
Failure-case analysis	Examine misplots and unstable cases manually	High	Errors can be diagnosed in terms of extraction, scoring, or theory mismatch rather than unexplained magic.

10 Recommended study design

A realistic research program would run in four passes.

10.1 Pass 1: formal benchmark construction

Build a benchmark set of short texts with adjudicated gold extractions. The set should cover clear cases, mixed cases, adversarial cases, rhetorical cases, and narrow topic fragments. This pass is mainly about extractor selection and ontology cleanup.

10.2 Pass 2: human agreement and profiler stress-testing

Have multiple human raters annotate the same texts under the contract categories. Then compare both humans and LLMs against adjudicated outputs. Stress-test the profiler by varying weights and disabling tie-break heuristics to see where plots are robust and where they are fragile.

10.3 Pass 3: longitudinal datasets

Collect sequences of texts from the same source over time. These can be diary-like reflections, debate transcripts, public essays, or simulated curricula. The aim is to see whether plotted movement is interpretable, whether gates evolve coherently, and whether canon memory improves rather than distorts later interpretation.

10.4 Pass 4: predictive studies

Use downstream tasks such as contradiction resolution, revision under counterevidence, or coherence under pressure. If the profiler is valid, its outputs should predict something about these tasks beyond superficial sentiment or topic classification.

11 Design recommendations

Several changes would improve scientific sharpness without changing the graph theory.

1. Freeze a public benchmark before tuning weights too aggressively.

2. Separate lexical tie-break features from the main score so they can be disabled and compared in ablation studies.
3. Log extractor uncertainty more explicitly rather than relying only on confidence fields embedded in events.
4. Add a profile-update threshold so tiny fragments do not over-write canon memory too easily.
5. Distinguish clearly between temporary situational emphasis and durable philosophical disposition in future profiler versions.
6. Keep avatar semantics fully outside validation claims.

12 Overall assessment

The core idea here is strong. The repository is doing something more serious than visual metaphor and more disciplined than direct LLM scoring. The graph theory is clear. The pipeline separates extraction from scoring. The profiler exposes its formulas. The visualizer preserves geometric integrity. Those are all real advantages.

The weak point is also clear. The current measurement layer is still a first-generation operationalization. The gate set may change. The signal weights may change. The tie-break heuristics may change. The benchmark does not yet appear to be frozen and published. None of that invalidates the project. It simply means the correct claim right now is that this is a scientifically promising candidate instrument, not a finished validated instrument.

That is still a strong position. Many systems in this space never become auditable enough to test seriously. This one already is.

13 Conclusion

Taken as a whole, Philosopher's Stone is best understood as a companion system around the Epistemic Octahedronrather than merely a graph viewer. The graph paper supplies the final geometric and semantic core. The codebase supplies a structured pipeline for turning text into auditable evidence, deterministic semantic aggregation, and geometrically valid visualization.

The right scientific framing is straightforward. The graph can be treated as final theory for the project. The contract and profiler can be treated as a provisional instrument specification. The next serious step is not rhetorical polishing. It is benchmark construction, extractor selection, inter-rater agreement, longitudinal testing, and predictive validation.

If that work is done well, the project could become a rare thing: a worldview-modeling system whose normative theory is explicit, whose measurement logic is inspectable, and whose claims can actually be tested.

A Quick reference: final versus provisional components

Category	Status
Octahedral graph geometry	Final theoretical core
Axis semantics	Final theoretical core
Origin and vertex meanings	Final theoretical core
Projection rule	Final theoretical core with deterministic implementation
LLM contract wording	Provisional operationalization
Gate set and gate weights	Provisional operationalization
Signal ontology and signal weights	Provisional operationalization
Tie-break keywords	Provisional fallback heuristic
Canon-memory rules	Provisional operationalization
Avatar mapping	Interface only
Visualizer surface-lock behavior	Implementation of final geometry